



Assessing the Impact of Offshore Wind Power Deployment on Fishery: A Synthetic Control Approach

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Abstract

Global efforts to decarbonize the energy sector are necessary to curb global warming, and many countries plan to achieve this through the rapid deployment of offshore wind power. However, the deployment will likely decelerate if local fishers plying their trade near maritime areas likely to host offshore wind farms oppose the deployments due to anticipated fishery disruptions. To date, there is no body of knowledge on the causal impact of offshore wind farm installation on local fisheries. Using fishery production panel data at the municipality level in Japan, this study applies a synthetic control method to measure causal impacts. The results suggest that offshore wind farms currently installed in Japan are unlikely to disrupt local fisheries. We find no statistically significant effects either on aggregated fishery production or on production categorized by fishery type. Moreover, we find no spatial spillover effects. Although the generalization of our findings requires caution because they are based on small-scale wind farms, our results imply that such moderate-size wind projects may not harm local fisheries.

Keywords Causal impacts · Observational data · Quantitative analysis · Renewable energy · Synthetic control method

JEL Classification R14 · Q22 · Q42 · Q51

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Abbreviations

SCM	Synthetic control method
ASCM	Augmented SCM
BACI	Before-after-control impact
BAG	Before-after gradient
METI	Minister of Economy, Trade, and Industry
GGS	Green Growth Strategy
PMRE	Act on Promoting the Utilization of Sea Areas for the Development of Marine Renewable Energy Power Generation Facilities
NEDO	New Energy and Industrial Technology Development Organization
DID	Difference in difference
SMFP	Statistics on Marine Fishery Production
SST	Sea surface temperature

1 Introduction

The deployment of renewable energy, such as solar and wind energy, is indispensable in achieving carbon neutrality and curbing global warming. The Intergovernmental Panel on Climate Change (IPCC) Special Report, Global Warming of 1.5 °C, states that global anthropogenic CO₂ emissions have to be at net-zero around 2050 to maintain the global temperature increase below 1.5 °C (IPCC 2018). Considering that energy usage is responsible for a large portion of anthropogenic CO₂, the decarbonization of the energy sector is necessary for carbon neutrality. The International Energy Agency's (IEA) scenario for realizing net-zero CO₂ emissions by 2050 assumes an increase of the global renewable energy capacity from 2990 GW in 2020 to 10,300 GW in 2030 and 26,600 GW in 2050 (IEA 2021). Specifically, the ratio of solar and wind electricity to the total generated electricity is expected to increase from 9% in 2020 to 40% in 2030, and 68% in 2050. Offshore wind power will significantly contribute to this increase in many regions and countries, including the European Union (EU), China, the United States (US), Korea, India, and Japan (IEA 2019).

To accelerate the deployment of renewable energy, policymakers need to fully understand the costs and benefits from both the economic and environmental viewpoints as renewable energy's facilities interact with both society and the surrounding environment (Borenstein 2012; Cullen 2013). Prior studies have investigated the economic and ecological impacts of well-deployed renewable energies¹ such as solar (Turney and Fthenakis 2011; Hernandez et al. 2014; Aman et al. 2015; Wee 2016; Tawalbeh et al. 2020) and onshore wind (Heintzelman and Tuttle 2012; Jensen et al. 2014; Mattmann et al. 2016; Zerrahn 2017; Jarvis 2021), and, with the increasing attention on offshore wind power, a growing body of literature has also investigated its ecological impacts (Boehlert and Gill 2010; Mangi 2013; Bailey et al. 2014; Premalatha et al. 2014). However, the economic impacts of offshore wind power are still poorly understood, except for Jensen et al. (2018),

¹ The ratio of solar and onshore wind energies capacity to the total capacity of all renewable energies was 50.48% in 2020 (IREA 2021). Moreover, these two energies occupy 88.95% of the annual capacity increase since 2019. For offshore wind energy, these numbers are 1.23% and 2.31%, respectively, implying that offshore wind power is still at its developing stage. See Gasparatos et al. (2017) for a systematic review on the ecological impacts of six renewable energy sources: solar, wind, hydro, ocean, geothermal, and bioenergy.

who investigated its effects on property values and found a null effect. Moreover, predicting the economic impacts based solely on ecological literature is infeasible if ecological impacts and their consequences on the economy are not fully understood. Thus, this study contributes to the literature by providing new evidence on the economic impacts, focusing on local fisheries.

The deployment of offshore wind power can decelerate if local fishers in maritime areas expected to host offshore wind farms oppose project deployments. Local fishers often worry about fishery income decreases caused by the construction, operation, and management of power facilities. The negative impacts on fishery income can arise from ecosystem changes and fleet dynamics (Yates and Bradshaw 2018). The typical impact pathways of ecosystem changes include noise disturbance, collision risk, and electromagnetic field emissions during the construction or operation phases (Bailey et al. 2014; Yates and Bradshaw 2018). If ecosystem changes result in a decrease in fish populations or fish species, fishery production will decline.² Fleet dynamics refer to changes in the fishing capacity of vessels and fishing activity; they include changes in the intensity of fishery efforts and their time and space allocation (Putten et al. 2012). For example, if power facilities are constructed in locations where fishers intensively conduct fishing activities, they have to find alternative fishing spots or change fishery types.³

Despite these concerns, studies assessing their effects on fisheries are limited. Alexander et al. (2013), Hooper et al. (2015), and Reilly et al. (2015) investigated fishers' attitudes toward offshore wind power deployment through interviews or surveys. Smythe et al. (2021) studied the attitudes of recreational anglers in places where an offshore wind farm had already been established. Although these studies help policymakers better understand potential conflicts between fishers and developers of offshore wind power projects, they do not measure the causal impacts of establishing power facilities. The lack of information on the causal economic impacts on fisheries prevents policymakers from implementing comprehensive cost-benefit analyses for offshore wind power projects (Snyder and Kaiser 2009), making it difficult to decide whether to promote them. Moreover, the deployment will decelerate if offshore wind power developers anticipate future conflicts with fishers and hesitate to invest in offshore wind power projects.

This study fills the gap in the literature by providing evidence regarding the causal impacts of offshore wind power deployment on local fisheries. We use publicly available municipality-level panel data on fishery production combined with offshore wind power deployment information from three municipalities in Japan. In a country where the developable land is small and surrounded by the sea, offshore wind power plays a critical role in the energy sector's decarbonization. Although the capacity of offshore wind farms has been minimal, the government has set a goal of attaining a 10 GW capacity by 2030 and a 30–45 GW capacity by 2040. The rapid deployment of offshore wind power will be challenging if perceived to negatively impact the local economy, surrounding environments, or both. One such concern of late is the negative impacts on fisheries, one of Japan's essential primary industries supporting about 200,000 fishery-related workers and maintaining the Japanese food culture. The Japanese government and offshore wind farm developers

² Ecosystem changes do not necessarily impact fish production or fish species negatively. Offshore wind farms are said to function as artificial reefs, contributing to the increase in fishery production nearby (Bailey et al. 2014).

³ These two channels are not exclusive of each other. For example, fleet dynamics can be observed after ecosystem changes.

try to avoid conflicts with fishers concerned about the possible fisheries disruption. Thus, the causal impact of installation on local fisheries is currently of great political interest in Japan.

This study assesses the causal impacts using the synthetic control method (SCM) developed by Abadie and Gardeazabal (2003) and Abadie et al. (2010, 2015). The SCM measures causal impacts as the difference between the fishery production of a treated municipality (i.e., where an offshore wind farm has already been installed) during the post-treatment period and that of a synthetic municipality, which is calculated by weighting municipalities without the treatment, namely, controlled municipalities (i.e., where no offshore wind farm has been installed). If the weights deliver a good approximation of fishery production in the treated municipality during the pre-treatment period, the synthetic control recovers fishery production during the post-treatment period that would be realized if the wind farm was not installed.⁴ The SCM is suited to comparative case studies, like the current one, compared to a difference-in-difference (DID) estimator, if the central assumption of common trend required for the DID estimator is violated due to non-random treatment assignment. In our context, locations for offshore wind farms are not randomly selected, and therefore, the common trend assumption might not be satisfied. The SCM instead allows us to control for factors that affect the endogenous selection by balancing the pre-treatment outcome of a treated municipality with that of controlled municipalities. Moreover, the SCM enables us to estimate time-varying effects, detecting the effects after years of installation. Since its development, the SCM has been broadly applied in empirical economics, including in the context of nuclear power deployment (Ando 2015) and renewable energy policies (Kirkpatrick and Bennear 2014; Maguire and Munasib 2016; Upton Jr et al 2017).⁵ In this study, we apply the augmented SCM (ASCM), which was recently proposed in Ben-Michael et al. (2021), to better approximate the pre-treatment trend of a treated municipality.

After running various estimation models with different outcomes, such as fishery types and fish species, and with and without spatial spillover effects, we found no statistically significant effects on fishery production. Hence, this implies that the offshore wind farms that are presently deployed might not necessarily disrupt fisheries. Moreover, we investigated the deployment impact on local per capita taxable income and found weak to no evidence of positive economic effects.

The current study contributes to the growing literature that investigates the impacts of offshore wind power deployment. Past studies mainly focus on ecological impacts (Boehlert and Gill 2010; Mangi 2013; Bailey et al. 2014; Premalatha et al. 2014), and only Jensen et al. (2018) investigated economic impacts on property values. This study is the first to investigate economic impacts with a special focus on fishery production. By doing so, we contribute to the vast literature that measures economic impacts of renewable energy installation (Aman et al. 2015; Wee 2016; Jensen et al. 2014; Zerrahn 2017; Jarvis 2021), providing new evidence on economic impacts of offshore wind power deployment. In addition, our study contributes to an ongoing discussion on the better use of coastal regions and marine and maritime areas and coordination between fishery and marine renewable

⁴ For the treated municipalities, the state without it is counterfactual. Therefore, fishery production after establishment in this counterfactual state is never observable. In the causal inference literature terminology, the *potential outcome* without the treatment is unobservable for controlled municipalities (Rubin 1974).

⁵ The SCM framework differs from impact evaluation methods often used in the ecological literature, such as the before-after-control-impact (BACI) and before-after-gradient (BAG) designs, in that the SCM weights multiple controlled units for comparison to the treated municipality (Methratta 2020).

energy facilities (Koundouri 2017; Yates and Bradshaw 2018; Koundouri 2021) by providing quantitative evidence on economic impacts of offshore wind power deployment.

The remainder of this study is organized as follows: Sect. 2 introduces the current state of offshore wind power and fisheries in Japan. Section 3 presents an overview of SCM. Section 4 describes the data and their sources. Section 5 presents the estimation results of the ASCM, and Sect. 6 discusses the results. Section 7 presents the conclusions.

2 Offshore Wind Power and Fishery in Japan

2.1 Offshore Wind Power in Japan

In October 2020, Japan's Prime Minister, Yoshihide Suga, declared to the realization of carbon neutrality by 2050. After the declaration, the Minister of Economy, Trade, and Industry (METI) formulated the Green Growth Strategy Through Achieving Carbon Neutrality in 2050 (GGS), an industrial strategy to achieve carbon neutrality by 2050 and create a virtuous cycle in the economy and environment (METI 2021). The strategy identifies offshore wind power deployment as one of the 14 essential pathways to achieve its goal and plans of attaining a capacity of 10 GW by 2030 and 30–45 GW by 2040. To drastically accelerate the deployment of offshore wind power, the Japanese government set an ambitious goal of reducing the electricity generation costs of offshore wind power to 8–9 JPY/kWh, which is a quarter of the current costs, by 2035.⁶ Moreover, in 2019, the government of Japan implemented the Act on Promoting the Utilization of Sea Areas for the Development of Marine Renewable Energy Power Generation Facilities (PMRE) to set rules for the occupancy of maritime areas and provide opportunities to involve stakeholders.⁷ Under the PMRE, municipalities take the initiative to designate maritime areas as promotion zones, where offshore wind farms will be installed in a way that enhances coordination between offshore wind power developers and preceding users such as fishers.⁸ These national and municipal efforts are expected to rapidly accelerate offshore wind power deployment in the following decades.

Before the PMRE was implemented, offshore wind farms were already installed in four different municipalities: Naraha, Choshi, Kitakyushu, and Goto. These farms were installed for verification tests and their results have been utilized to design national offshore wind power policies. This study focuses on offshore wind farms in three municipalities: Choshi, Kitakyushu, and Goto.⁹ These municipalities were selected based on geographical

⁶ In Europe, offshore wind power projects whose procurement costs are less than 10 JPY/kWh are becoming common. However, the same technology as Europe is difficult to deploy in Asia due to the difference in natural and geological conditions. Hence, the New Energy and Industrial Technology Development Organization (NEDO) made an original roadmap for technology development to deploy offshore wind farms in Asia with low costs (NEDO 2021). It promotes the development of offshore wind measurement methods appropriate to Asian natural conditions, designs of wind farms that are tolerant to natural disasters such as typhoons and earthquakes, and reduction of operation and management costs through smart maintenance that utilizes digital technologies and artificial intelligence.

⁷ See <http://www.japaneselawtranslation.go.jp/law/detail/?id=3519&vm=04&re=02>.

⁸ Offshore wind farms have not yet been installed under the PMRE at the time of writing this paper.

⁹ The excluded municipality, Naraha, is located in Fukushima prefecture. We decided to exclude the municipality as it is difficult to distinguish the impacts of offshore wind power deployment from those of the Great East Japan Earthquake in 2011. Other offshore wind farms are removed due to data unavailability.

conditions, such as wind speed, wave height, and possible environmental impacts. Table 1 summarizes the information of the three farms, and Fig. 1 shows their locations.

As Table 1 shows, offshore wind farms in this study are small-scale and have a capacity of approximately 2 MW. The distance from shore is relatively close: the maximum distance is five kilometers in Goto, whose farm is a floating type. The onset of construction was recent, with the oldest being in Choshi in 2010. Offshore wind power deployment can affect fisheries during the construction and operation phases. Therefore, we are interested in the impacts since the year of construction onset rather than the year of operation start.

In summary, offshore wind power in Japan is still developing, although the situation is likely to change dramatically in the coming decades. Four maritime areas have already been designated as promotion zones (shaded zones in Fig. 1), with a plan of installing offshore wind farms with a maximum capacity of 1.5 GW in total. Some fishers in Japan oppose such development projects, arguing that large-scale offshore wind farms could disrupt fisheries and reduce their incomes, and some protest movements have eventually turned into court cases.¹⁰ To date, the impacts of offshore wind power deployment, regardless of their scale, on local fisheries are not fully understood. Even in the case of small-scale offshore wind farms, negative impacts can occur. Therefore, it is natural to assume that these impacts may also exist for large-scale farms, which means that specific measures need to be undertaken to mitigate negative impacts. The first step would be analyzing small-scale farms to understand the impacts of offshore wind power deployment.

2.2 Japanese Fishery

In Japan, where the sea surrounds the land, fisheries are an essential industry. Fish flesh nourishes many people: the per capita supply of fish and fishery productions in 2017 was 45.8 kg, more than twice the world average—20.3 kg (FAO 2020). Although below its 1980s peak, fish capture production, including crustaceans and mollusks, was 3.13 million tons in 2018, the eighth largest globally. The fisheries industry still supported 162,249 and 40,181 fishers and fish farmers, respectively, in 2018; thus, fisheries have contributed to the development and sustenance of coastal communities.^{11,12}

Japan's total fishery production mainly comprises marine aquaculture and three other types of fisheries: coastal, offshore, and distant water fisheries. The three fisheries differ in the days of trips and fish species harvested.¹³ Offshore fisheries make the largest share of the total fishery production, followed by marine aquaculture and coastal fisheries (see Fig. 2). However, the production from coastal fisheries shows a slight decrease over time, although the decrease has been much smaller than that of distant fisheries since the mid-1970s, after the introduction

¹⁰ Fishers' opposition against offshore wind power deployment is not a unique problem in Japan and is observed in other regions and/or countries such as the US (<https://www.nationalfisherman.com/seafoodsou/rce/lawsuit-seeks-to-block-vineyard-wind-faults-nmfs-biological-opinion/>), Norway (<https://www.thelocal.no/20210326/why-norwegian-fishermen-are-against-more-offshore-wind-farms/>) and the EU (<https://thefishingdaily.com/latest-news/eu-fishing-industry-responds-to-the-offshore-renewable-energy-strategy/>).

¹¹ Coastal communities here refer to communities consisting of more than three fishers who live near a fishery harbor. The number of coastal communities in Japan was 6298 in 2013 (MAFF 2013).

¹² Fisheries are also of cultural importance in Japan. The FAO designated two coastal communities as the Globally Important Agricultural Heritage Systems (GIAHS) for their uniqueness and sustainability. For details on GIAHS, see <http://www.fao.org/giahs/giahsaroundtheworld/en/>.

¹³ Coastal, offshore, and distant fisheries require a single day, several days, and more than a couple of days for trips, respectively.

Table 1 Offshore wind farms in treated municipalities

	Choshi	Kitakyushu	Goto
Capacity (MW)	2.4	2.0	2.0
Distance from shore (km)	3.0	1.4	5.0
Type	Fixed-bottom	Fixed-bottom	Floating
Construction start year	2010	2012	2014
Operation start year	2013	2013	2015

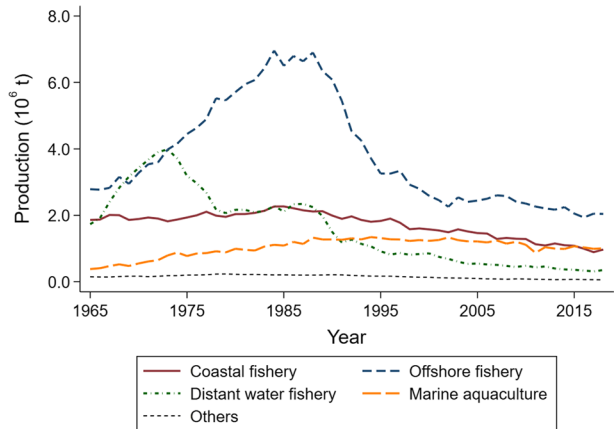
Fig. 1 Location of offshore wind farms and promotion zones

of 200 nautical-mile zones, and offshore fisheries, whose production is considered to decline due to the decrease in the sardine population, since the early 1990s.

The installation of offshore wind farms may have a differential impact on these fishery types. For example, fishing areas of coastal fisheries are more likely to overlap with maritime areas where offshore wind farms are installed than offshore and distant water fisheries, whose vessels usually travel far away. Thus, in our analysis in Sect. 5, we separately measure the impacts on coastal and non-coastal fisheries.

In contrast to the fisheries trends, marine aquaculture production has been stable during the same period. Marine aquaculture in Japan interacts intensively with marine fisheries; they often share the same maritime areas, and marine fisheries provide feed and juvenile fish to marine aquaculture. Therefore, if offshore wind power deployment disrupts marine fisheries, marine aquaculture could also be affected. We present the estimation results for marine aquacultural production in Appendices B and C.

Fig. 2 Fishery production in Japan. Notes: This figure is based on the Statistics on Marine Fishery Production (see Sect. 4 for data details). “Others” include inland-water fishery and inland-water aquaculture



3 Methodology

3.1 Synthetic Control Method

Suppose that there are $N + 1$ municipalities indexed by $i \in \{1, 2, \dots, N + 1\}$ and data span T periods indexed by $t \in \{1, \dots, T_0 + 1, \dots, T\}$, where $T_0 + 1$ denotes the year of installation of an offshore wind farm ($0 < T_0 < T - 1$). For ease of description, we assume that there is only one municipality in which a power facility is established and that the first unit ($i = 1$) corresponds to this municipality. The parameter of interest in this study was the causal impact of offshore wind power deployment on local fisheries. Let τ_{1t} denote the impact in year t ($t > T_0$). Let $Y_{1t}(1)$ and $Y_{1t}(0)$ denote the potential outcomes (i.e., fishery production realized) with and without the treatment, respectively. Then, for a post-treatment period $t > T_0$, the parameter of interest τ_{1t} can be defined as follows:

$$\tau_{1t} = Y_{1t}(1) - Y_{1t}(0). \quad (1)$$

Let Y_{1t} be the realized outcome. Because an offshore wind farm is installed in municipality 1, $Y_{1t}(1)$ is observable and $Y_{1t}(1) = Y_{1t}$ for $t > T_0$. However, the fishery production that would be realized if offshore wind power was not deployed, $Y_{1t}(0)$, is unobservable for $t > T_0$. Therefore, we cannot directly obtain τ_{1t} .

If there exists a municipality ($i = \ell$) where offshore wind power has not been deployed and whose characteristics, including fishery states, perfectly match those of municipality 1, we can substitute $Y_{1t}(0)$ with $Y_{\ell t}$ and obtain τ_{1t} . However, in the real world, it is not feasible to find such municipalities. Instead of finding a single municipality, the synthetic control method (SCM) weights controlled municipalities to approximate municipality 1 (Abadie and Gardeazabal 2003; Abadie et al. 2010, 2015). Since its development, SCM has been applied in many empirical work contexts (Abadie 2021) and is now seen as “arguably the most important innovation in the policy evaluation literature in the last 15 years” (Athey and Imbens 2017). The fields of application include impact evaluation of nuclear power facilities on the local economy (Ando 2015) and renewable energy policies on its deployment (Kirkpatrick and Benneer 2014; Maguire and Munasib 2016; Upton Jr et al 2017), although there is no application to offshore wind power.

SCM delivers the predictions of $Y_{1t}(0)$ during the post-treatment period ($t > T_0$) by weighting N controlled municipalities ($i = 2, 3, \dots, N+1$). The weights are computed in a way that minimizes the difference in characteristics between Municipality 1 and the weighted municipalities during the pre-treatment period ($t \leq T_0$). These characteristics may include pre-treatment fishery outcomes. With a set of weights, $\mathbf{w} = (w_2, \dots, w_{N+1})'$, at hand, the synthetic control estimator of $Y_{1t}(0)$ is given as

$$\hat{Y}_{1t}(0) = \sum_{i=2}^{N+1} w_i Y_{it}. \quad (2)$$

Then, the synthetic control estimator of τ_{1t} is:

$$\hat{\tau}_{1t} = Y_{1t} - \hat{Y}_{1t}(0).$$

The weights are restricted to lie between zero and one and sum to one to avoid extrapolation, which may make inferences unreliable (King and Zeng 2006). These restrictions help researchers avoid putting too large weights on municipalities that are not similar to Municipality 1. See Appendix A for the methodological details of SCM.

3.2 Augmented SCM

The credibility of the synthetic control estimators critically depends on whether the weights provide an excellent approximation of Municipality 1's pre-treatment outcomes. If fishery outcomes during the pre-treatment period balance poorly between Municipality 1 and the synthetic municipality, the weighted outcome cannot be a good approximation of the unobservable post-treatment outcome of Municipality 1. It is often the case in applications, however, that such a perfect pre-treatment fit is infeasible. Ben-Michael et al. (2021) developed an augmented synthetic control method (ASCM) to correct the possible biases introduced by imperfect pre-treatment approximation. In ASCM, modest extrapolation is allowed under certain penalties for extrapolation. The ASCM estimator of $Y_{1t}(0)$ is constructed to correct for possible biases as follows:

$$\hat{Y}_{1t}^{\text{ascm}}(0) = \sum_{i=2}^{N+1} w_i Y_{it} + \left(\mathbf{X}_1 - \sum_{i=2}^{N+1} w_i \mathbf{X}_i \right) \boldsymbol{\eta}_x + \left(\mathbf{Z}_1 - \sum_{i=2}^{N+1} w_i \mathbf{Z}_i \right) \boldsymbol{\eta}_z, \quad (3)$$

where $\mathbf{X}_i$ is a vector of pre-treatment fishery outcomes, $\mathbf{Z}_i$ is a vector of municipality characteristics that are not influenced by the treatment, and $\boldsymbol{\eta}_x$ and $\boldsymbol{\eta}_z$ are vectors of additional weights. Unlike conventional SCM, w_i can take negative values. Since a perfect pre-treatment fit may fail, we apply ASCM to estimate $Y_{1t}(0)$ and τ_{1t} .¹⁴ See "Appendix A" for the ASCM's methodological details.

The SCM approach requires researchers to specify a donor pool, a set of controlled municipalities used for the pre-treatment approximation. We define a donor pool for a treated municipality as the set of municipalities where offshore wind power had not been deployed as of

¹⁴ While the large literature exists on the application of the SCM in environmental and resource economics, there is yet little application of the ASCM except for Amador-Jiménez et al. (2020) and Cole et al. (2020). Note that pre-treatment trends might poorly balance even when we apply the ASCM. In this case, causal inferences fail, and we cannot derive any arguments.

2018 and whose average fishery production during the pre-treatment period is in the same 10th quantile as the treated municipality. That is, municipalities in the donor pool are similar to the treated municipalities in terms of pre-treatment fishery production. We form donor pools each time we change the outcomes.

Note that a perfect pre-treatment fit does not guarantee the exact prediction of the counterfactual outcome. For example, municipalities in the donor pool may receive large weights even when they differ from a corresponding treated municipality in terms of observables, unobservables, or both. Since we cannot directly match unobservables, we include covariates expected to affect fishery production in our models (see Sect. 4 for details of the covariates). In Sect. 5, we discuss how well our models balance the covariates, as well as fishery outcomes. In the same section, we also present the results suggesting that including covariates improves the outcome's pre-treatment balances, but excluding them has minor effects on results.

3.3 Comparison to Standard DID Estimator

In the program evaluation literature, it is common to estimate treatment effects through a DID estimator using the following regression equation:

$$Y_{it} = \alpha_i + \tau \cdot \mathbb{1}[i = 1] \cdot \mathbb{1}[t > T_0] + \lambda_t + \varepsilon_{it}, \quad (4)$$

where $\mathbb{1}[\cdot]$ is an indicator function that takes the value of 1 if the argument inside the bracket is true and 0 otherwise, α_i captures municipality-fixed effects, λ_t represents time-fixed effects, and ε_{it} is an unobserved error term. However, considering that the sites of offshore wind farms are not randomly assigned, the use of the standard DID estimator is inappropriate in the current study. Site selection based on geographical and environmental conditions may violate the common trend assumption: the central assumption required for the DID approach is that treated municipalities' outcomes would trend similar to those of controlled municipalities in the absence of the treatment. Moreover, the comparison between Eqs. (1) and (4) indicates that the SCM allows for time-varying effects, while the standard DID does not. This distinction is notable in this study, as changes in the ecosystem and fishers' behavior after installing offshore wind farms can affect fishery production differently over time.

Nonetheless, we estimate Eq. (4) and present the results in Tables 6 and 7 of Appendix C. As discussed in Sect. 5, the estimated effects and implications derived from the SCM estimator are partially different from those derived from the DID estimator. This difference highlights the importance of considering the influences of site selection and non-constant impacts over time.

4 Data

4.1 Data Source

Information on fishery production was obtained from the Statistics on Marine Fishery Production (SMFP).¹⁵ The SMFP records each municipality's annual marine fishery

¹⁵ Source is: <https://www.e-stat.go.jp/en/statistics/00500216>.

production from 2004 to 2018 at the fishery-type and fish-species levels.¹⁶ It also records annual aquacultural production at the municipal level. All landing facilities, which consist of fishery cooperatives and fishery companies, and all fishers¹⁷ who do not belong to landing facilities are asked to report the amount of fishery production by fish species and fishery types. Production in a municipality is the sum of fish caught by fishers living in the municipality. Therefore, we utilize fishery production as our outcome of interest.

It is worth emphasizing SMFP measures. Production in a municipality (say A) recorded by the SMFP is the amount of fish landed by fishers who live in A. In other words, fish caught by those who catch fish near A and take it to one of A's landing facilities but live in another municipality (say B) is recorded as B's production. The gap between the actual amount of fish caught in A's maritime areas and its production recorded by the SMFP is expected to increase for non-coastal fisheries that take several days to fishing grounds. An estimation that ignores this discrepancy might not capture the true effects of installing offshore wind farms on fisheries. Thus, in this study, we first start our estimation from the total fishery production, which does not consider the gap, and then, as our preferred model, we focus on coastal fisheries, whose gap is expected to be smaller than non-coastal fisheries such as offshore and distant water fisheries.¹⁸

To approximate pre-treatment trends of fishery outcomes of treated municipalities, we collected fishery-related, socio-economic, and environmental covariates at the municipal and prefectural levels. Fishery-related covariates include the number of fishers, the number of fishery facilities, and the annual fishery production per fishery cooperative. These covariates are provided at the municipal level, except for the last one. Socio-economic characteristics include the number of workers in the primary sector, taxable income per capita, population, and the share of the elderly, all of which are recorded at the municipal level. Environmental covariates are the potentials of wind power generation measured by wind strength, mean sea surface temperature (SST), and the number of storm days. The potential data represent the average wind strength within three kilometers of each municipality's seashore. SST was measured at 122 observation points. Each municipality was matched with its nearest point. A storm day is defined as a day when the maximum wave height exceeds four meters. The wave height was observed hourly at 207 observation points, and we matched the nearest point to each municipality. We observed these environmental covariates at the municipal level. See Table 2 for the variables we collected with their units, data periods, and sources.¹⁹

We expect these variables to better approximate the pre-treatment trend. In fact, the pooled ordinary least squares regression, wherein we regress fishery production on these characteristics with year and prefectural fixed effects, generates an adjusted R^2 of 0.65 and 0.43 for the total and coastal fisheries, respectively, suggesting that these variables capture a large part of the variation in fishery production (see Tables 4 and 5 of Appendix C for

¹⁶ Fishery production at the prefectural level has been available since 1956 and is used to draw Fig. 2.

¹⁷ Fishers here refer to households or business facilities that harvest marine animals and plants or engage in aquaculture to sell their productions for a living.

¹⁸ One way to remove the gap is the use of fishers' logbooks that record the production by fishing ground (Abe and Anderson 2022). Unfortunately, no public, detailed information is available at the municipal level.

¹⁹ Source for Renewable Energy Potential System is, <http://www.renewable-energy-potential.env.go.jp/RenewableEnergy/index.html>. Source for Japan Oceanographic Data Center is, <https://www.jodc.go.jp/jodcweb/index.html>. Source for other control variables is, <https://www.e-stat.go.jp/en>.

detailed results). In addition to these variables, we include the share of fishery types to capture fishery characteristics in the SCM estimation.

4.2 Data Description

From the municipalities listed in each data source, we exclude those that do not face the sea and those whose fishery production is zero for the entire sample period (i.e., from 2004 to 2018). This process leaves 549 municipalities spanning 15 years, resulting in 8235 observations.²⁰ Since a donor pool is the set of controlled municipalities whose average fishery production during the pre-treatment period is in the same 10th quantile as the treated municipality, the average size of donor pools is around 55. Table 3 presents the descriptive statistics of the variables listed in Table 2—we separately aggregated municipalities with offshore wind farms and the rest of the municipalities over time.

Table 3 indicates that the fishery in Choshi and Goto is much more active than the average municipality, while the fishery in Kitakyushu is less active. A large portion of total production in Choshi comes from the non-coastal fishery category, and coastal fishery is the worst in all the three municipalities. Coastal fisheries are the most active in Goto. In terms of the number of fishers, fishery facilities, and fishery facility workers, Goto is the largest among the three. At the prefectural level, Chiba, where Choshi belongs, has the largest production per fishery cooperative. Figure 3 represents the trends of fishery production of treated municipalities, municipalities in a corresponding donor pool, and the average in all controlled municipalities.

In addition to fishery-related variables, Choshi had the largest number of workers in the primary sector. In contrast, the number in Kitakyushu was the smallest among the three. The taxable income per capita and population in Kitakyushu (Goto) is the largest (smallest) among the three treated municipalities, while the proportion of the elderly in Kitakyushu (Goto) is the smallest (largest). Thus, Kitakyushu captures the features of urban municipalities, while Goto captures the features of rural municipalities. The wind is the strongest in Kitakyushu, while that in Choshi and Goto is a little lower than average, which suggests that, based solely on the average wind strength, Kitakyushu is potentially the most suitable for offshore wind power deployment among the three municipalities. The mean SST in the treated municipalities was higher than the average, with Goto having the highest among the three. Goto is also the municipality with the most storm days among the three, although the number is still smaller than the average.

5 Results

5.1 Overall Impacts

Figure 4 represents the ASCM results for total fishery production, which aggregate the production of all fishery types and fish species; Fig. 4a–c represent Choshi, Kitakyushu, and Goto, respectively. The solid line represents the fishery production of a treated municipality, while the dashed line represents the production of its synthetic municipality. The

²⁰ Some municipalities disappeared from the list after a certain year due to municipal mergers. We regard merging municipalities as a single municipality if the merge information is available.

Table 2 Summary of variables

Variables	Year	Source
<i>Outcome</i>		
Fishery production ^a	2004–2018	Statistics on Marine Fishery Production
<i>Fishery-related covariates</i>		
# fishers ^a	2008, 2013, 2018	Fishery Census
# fishery facilities ^a	2009, 2014	Economic Census
# fishery facility workers ^a	2009, 2014	Economic Census
Production per cooperative ^b	2004–2018	Fishery Cooperative Statistics
<i>Socio-economic covariates</i>		
# primary sector workers ^a	2005, 2010, 2015	National Census
Taxable income per capita ^a	2004–2018	Prefectural Accounts, National Census
Population ^a	2005, 2010, 2015	National Census
Rate of age $\geq 65^a$	2005, 2010, 2015	National Census
<i>Environmental covariates</i>		
Wind power potential ^a	2019	Renewable Energy Potential System
Mean SST ^a	2004–2018	Japan Oceanographic Data Center
# storm days ^a	2004–2018	Japan Oceanographic Data Center

^{a,b}represent data at the municipal and prefectural levels, respectively

vertical dotted lines represent the year construction started, whereas the horizontal dotted lines correspond to zero. If two lines to the left-hand side of a vertical dotted line (i.e., during the pre-treatment period) coincide, the pre-treatment trend is balanced. The shaded area represents the 95% confidence interval of the treatment effects, where the baseline is a synthetic municipality's outcome.

Figure 4 shows that pre-treatment trends in fishery production are well approximated. Importantly, none of the municipalities exhibit statistically significant impacts on fishery production after deploying offshore wind power. The largest effects judged by their absolute value of Choshi, Kitakyushu, and Goto are -24.79 , 0.96 , and 8.23 with their corresponding p -values of 0.71 , 0.56 , and 0.55 , respectively. In addition, aggregating these effects over treated municipalities and years following the method proposed by Ben-Michael et al. (2021) delivers an average treatment effect of -2.05 with a p -value of 0.82 . Thus, our results suggest that the installation of offshore wind farms does not seem to disrupt fisheries.

The null effects on total fishery production are in line with those derived from DID regressions in Table 6 in Appendix C except for the one derived from a model in Kitakyushu without including covariates. Table 6 indicates that the coefficient is positive (0.288) and statistically significant at the 1% level. This discrepancy implies the importance of controlling for observables, the possible violation of the common trend assumption, and non-constant impacts over time, which validates our SCM approach.

5.2 Impacts by Fishery Types

The previous subsection presents the results based on total production as the first step of our estimation. Hereafter, we mainly focus on coastal fishery production, emphasizing that

Table 3 Summary statistics

Variables	Choshi		Kitakyushu		Goto		Others	
	Mean	S.D.	Mean	S.D.	Mean	S.D.	Mean	S.D.
<i>Outcomes</i>								
Total fishery production (10 ³ t)	55.02	6.82	3.53	0.30	15.62	1.81	6.72	15.69
Coastal fishery production (10 ³ t)	0.68	0.25	0.76	0.08	4.39	0.56	0.93	2.98
Fixed net	0.00	0.00	×	×	2.01	0.26	0.67	2.28
Seine net	0.03	0.02	×	×	×	×	0.34	1.67
Gill net	×	×	×	×	1.19	0.11	0.29	1.32
Shell & weed collecting	0.02	0.02	×	×	0.13	0.09	0.24	1.12
Fishing	0.63	0.22	0.52	0.04	0.60	0.12	0.06	0.16
Harling	×	×	0.24	0.06	0.46	0.19	0.03	0.24
Non-coastal fishery production (10 ³ t)	54.35	6.63	2.78	0.34	11.24	1.57	5.79	14.95
<i>Fishery-related covariates</i>								
# fishers	495.00	22.54	691.80	63.01	1347.20	181.61	341.20	441.30
# fishery facilities	117.30	18.05	34.00	0.00	278.30	14.15	72.65	130.80
# fishery facility workers	4.00	0.00	15.33	0.49	19.33	0.49	5.71	9.53
Production per cooperative (10 ³ t)	61.15	17.18	7.33	6.85	25.06	7.59	40.75	69.22
<i>Socio-economic covariates</i>								
# primary sector workers (10 ³)	2.92	0.58	2.40	0.56	2.09	0.15	1.54	1.82
Taxable income per capita (M JPY)	1.18	0.04	1.23	0.04	0.87	0.08	1.06	0.28
Population (10 ³)	70.59	4.40	979.40	13.48	41.40	3.13	85.10	238.80
Rate of age ≥ 65	0.29	0.03	0.25	0.03	0.33	0.03	0.29	0.07
<i>Environmental covariates</i>								
Wind power potential (m/s)	2.86	0.00	3.38	0.00	2.75	0.00	3.06	1.43
Mean SST (°C)	19.57	0.41	19.29	0.28	21.38	0.33	18.27	3.41
# storm days	0.00	0.00	0.00	0.00	25.73	7.45	38.20	96.66
Observation	15		15		15		8190	

The distinction between coastal and non-coastal fisheries is based on the MAFF. “×” means missing values due to data confidentiality. Due to these missing values, non-coastal fishery production can include coastal fishery production

this focus is crucial in estimating treatment effects for at least two reasons. First, as discussed in Sect. 4, estimation focusing only on the total fishery production might not capture true treatment effects because of the possible gap between the production recorded by the SMFP and the actual amount of fish captured. The fishing grounds of coastal fisheries are closer to landing facilities than those of non-coastal fisheries. Therefore, a focus on coastal fisheries is expected to minimize this gap. Second, the focus on coastal fisheries helps detect the heterogeneity of deployment impacts on fisheries. Offshore wind farms can have differential impacts on different types of fisheries, as discussed in Sect. 2. Table 1 suggests that offshore wind farms have been installed in coastal maritime areas, and thus, coastal fisheries can be exposed to larger impacts than distant water fisheries.

To measure treatment effects and shed light on the heterogeneity, we divide the overall fishery production into coastal and non-coastal fisheries and separately estimate the weights. The coastal fishery includes six types of fisheries, as shown in Table 3. Note that some data are inaccessible from available data sources owing to data confidentiality.

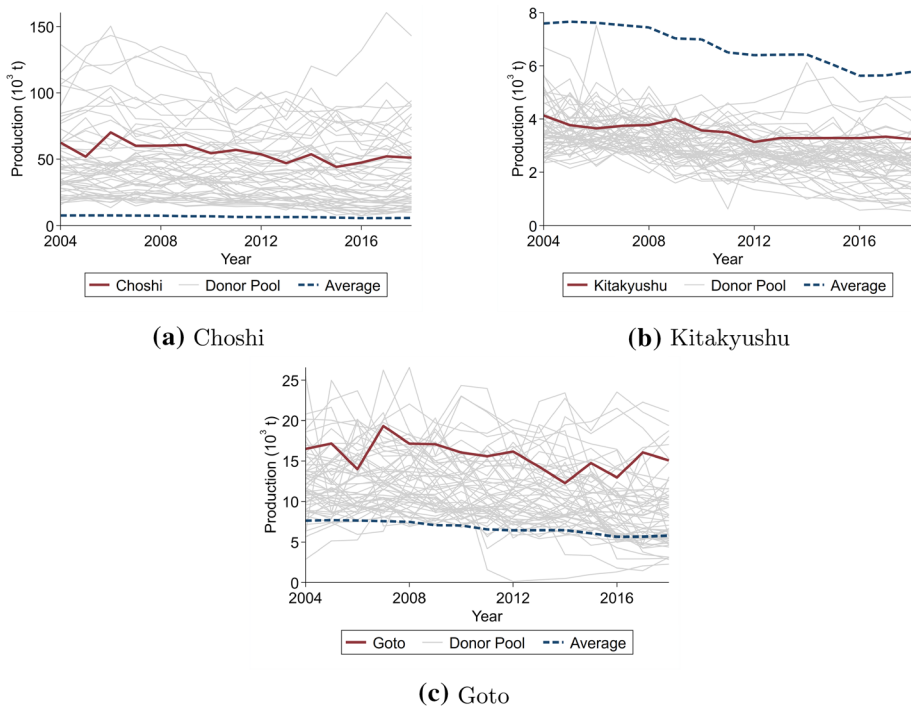


Fig. 3 Total fishery production over time

Therefore, coastal fishery values are under-reported, whereas those of non-coastal fisheries are over-reported. Note that we re-formed the donor pools for each model.

Figure 5 plots ASCM estimation results of coastal and non-coastal fisheries. Figure 5a–c represent the results of coastal fisheries, while Fig. 5d–f represent those of non-coastal fishery. The trends of fishery production during the pre-treatment periods balance quite well in all municipalities and fishery types. The models do not detect any impacts on coastal and non-coastal fisheries that are statistically distinguishable from zero. The largest effects on coastal fishery judged by their absolute value of Choshi, Kitakyushu, and Goto are -0.18 , 0.61 , and 0.95 with their corresponding p -value of 0.71 , 0.44 , and 0.73 , respectively. The average treatment effect on coastal fishery production aggregated over treated municipalities and years, which was computed by following Ben-Michael et al. (2021), is -0.08 with a p -value of 0.86 . Thus, offshore wind power deployment does not seem to have distinct impacts on fisheries.²¹ Perhaps surprisingly, this implication does not depend on the fishery scale, types of offshore wind farms (i.e., fixed-bottom or floating), and distance from shore (see Tables 1 and 3).

In contrast to the case of total fishery production, the null effects on coastal fishery production contradict those derived from DID regressions in Table 7 in Appendix C, where the estimates are statistically significant at a level of 10% at least. It is to be noted that the three treated municipalities were not randomly selected. Therefore, the common

²¹ In Appendix B, we estimate the ASCM weights for each fishery type. The results in Figs. 11, 12, and 13 suggest that offshore wind farms affect none of fishery types statistically.

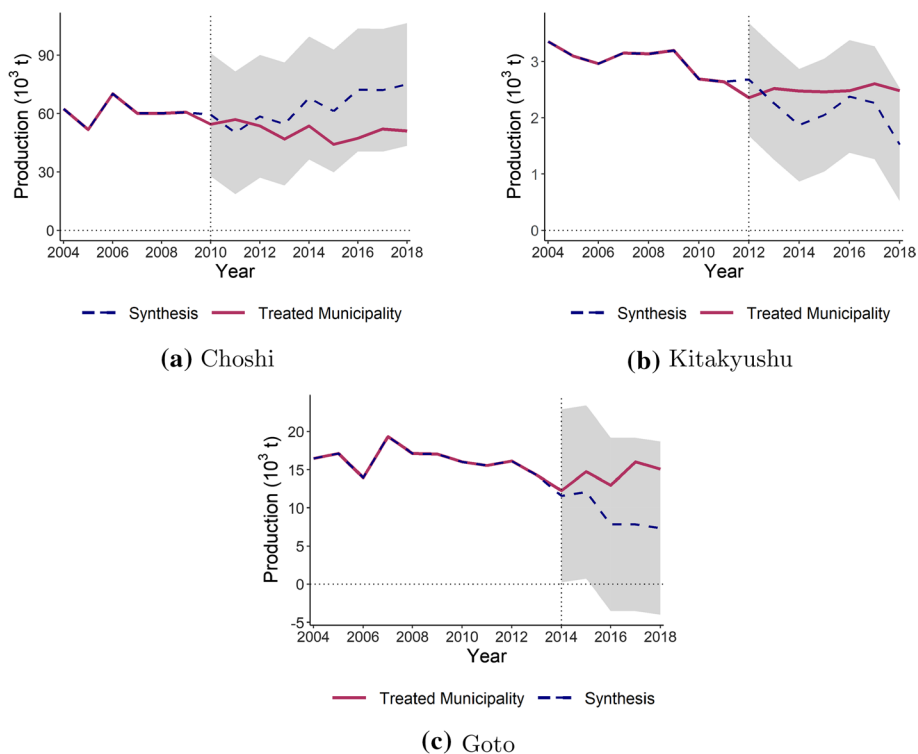


Fig. 4 ASCM estimation results for overall production. Notes: The shaded areas show 95% confidence intervals

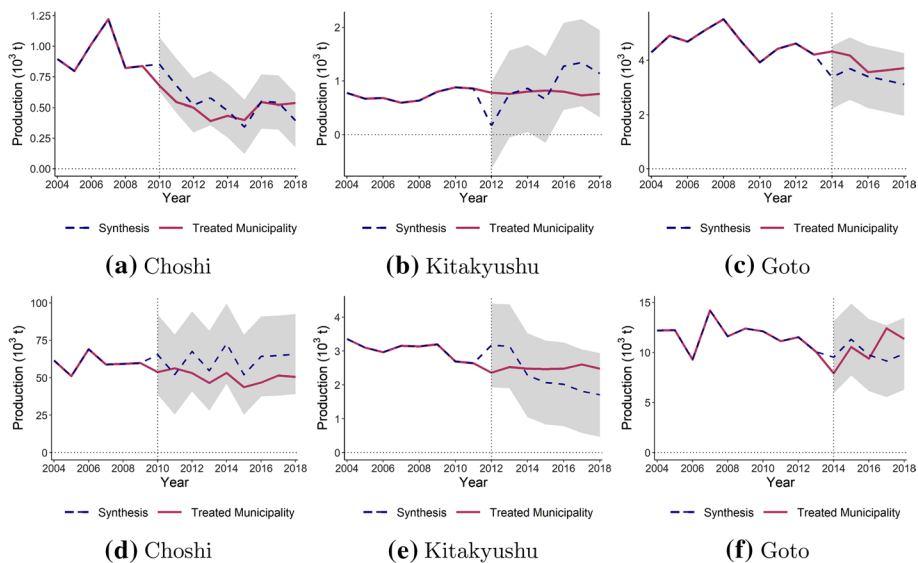


Fig. 5 ASCM estimation results for coastal and non-coastal fisheries production. Notes: The shaded areas show 95% confidence intervals. **a–c** show the results of the coastal fishery. **d–f** show the results of the non-coastal fishery

trend assumption is likely to be violated and DID estimators might be biased. The discrepancy between our ASCM results and DID results highlights the importance of capturing site selection and non-constant impacts over time, suggesting the validity of our SCM approach.

The excellent pre-treatment fit does not guarantee a perfect prediction of counterfactual outcomes. Even if pre-treatment outcomes balance well, as shown in Fig. 5, other observables, unobservables, or both can differ considerably. In such cases, synthetic post-treatment outcomes do not deliver valid counterfactuals. Therefore, to examine how well our models approximate variables other than total fishery production, in Table 10 of Appendix C, we compare the averages of covariates between treated municipalities, their corresponding syntheses, and all controlled municipalities. The table shows that our models approximate most of the variables well, supporting the validity of our counterfactual outcomes.

In Appendix B, we decompose the overall production based on the fish species and estimate the ASCM weights for each species. We focused only on the major 13 fish species in Japan.²² Overall, we find no statistically significant effects (see Figs. 14, 15, and 16; see also Table 8 for the relationship between fishery types and fish species in Japan). Moreover, in Appendix B, we consider the impacts on marine aquaculture, which intensively interacts with marine fisheries, as introduced in Sect. 2. The information on marine aquaculture is summarized in Table 9, and the estimation results are shown in Fig. 17. Choshi was eliminated from this analysis because data on aquaculture production were unavailable due to data confidentiality. The results indicate that offshore wind power deployment has little effect on marine aquaculture.

5.3 Spatial Spillover Effects

A further concern for our estimation results is the spatial spillover effects of offshore wind power deployment. Remember that fishery production in a municipality is the sum of fish caught by fishers living in the municipality. In other words, when a fisher living in Municipality A catches fish in a maritime area facing Municipality B, its production is recorded as Municipality A's production. Therefore, if fishers from neighboring municipalities catch fish near the power facility, an offshore wind farm can affect not only the municipality where it is installed but also neighboring municipalities. In addition, wind farms cause turbid wakes (Vanhellemont and Ruddick 2014), potentially influencing the marine ecosystem and fishery production in neighboring municipalities. If spatial spillover effects exist, our estimation results are biased because we include municipalities affected by offshore wind farms in the donor pools. To address the spatial spillover effects, we consider two procedures. First, for a municipality where an offshore wind power facility is established, we eliminate municipalities that belong to the same prefecture as the treated municipality and the neighboring prefectures. In other words, from donor pools, we eliminate municipalities that neighbor the treated municipality or its prefecture. If spatial spillover exists, this procedure might lead to an underestimation of the impact of interest. Hence, in the second procedure, we estimate weights by considering neighboring municipalities as sites where offshore wind power is deployed. In this procedure, we aggregate the municipalities into a

²² Major fish species satisfy at least one of the following conditions: (1) designated under the Total Allowable Catch scheme, which sets a species-specific maximum annual allowable catch, (2) fish production being more than 10 thousand tons and its production value being more than 10 billion JPY, or (3) a production value being more than 10 billion JPY and listed on the national resource management guideline.

single unit by taking the average fishery production.²³ To detect spatial spillover effects on neighboring municipalities, we focused on coastal fisheries. Figure 6 shows the results of investigating spillover effects.

Figure 6a–c present the results of the first procedure, where municipalities and prefectures with neighboring treated municipalities are eliminated from donor pools. In contrast, Fig. 6d–f represent the results of the second procedure, where neighboring municipalities are aggregated into a single treated unit. In line with the previous results, pre-treatment trends balance well, and the effects on fishery production are statistically indistinguishable from zero in all municipalities for both procedures. Specifically, the results delivered by the former procedure are similar to the results in Fig. 6, suggesting that municipalities neighboring a treated municipality did not receive large weights. Therefore, the size of the spillover effects seems to be limited.

5.4 Robustness Checks

This subsection first provides the estimation results of two additional analyses following Abadie (2021), which are used to examine the robustness of our estimation results. The first is a backdating analysis that investigates the possibility of anticipating the effects of offshore wind power deployment. The second is a leave-one-out analysis that tests the sensitivity of our results to changes in donor pools. Finally, we investigated how our results are robust to the exclusion of covariates. The outcome variable in this subsection is coastal fishery production.

5.4.1 Backdating Analysis

The year of treatment was set as the onset of the construction of offshore wind farms. If local fishers know the project years before the treatment and anticipate that fishery production will decrease after the construction, they might increase fishery efforts before the construction starts, enhancing fishery production during the pre-treatment period (McDermott et al. 2019). In contrast, if maritime exploration before the onset of construction affects the surrounding environment (Bailey et al. 2014), negative impacts may appear before the year of construction. Therefore, to investigate the possibility of anticipation effects, exploration effects, or combined effects, we regard the year of the treatment as one and two years before the actual treatment and re-estimate the ASCM weights.

The results are shown in Fig. 7. Figure 7a–c show the results of the one-year ahead analyses, whereas Fig. 7d–f show the results of the two-year ahead analyses. The pre-treatment trends balance well in all the models. None of the effects are statistically distinguishable from zero, which is inconsistent with the existence of anticipation effects, exploration effects, or both. Thus, offshore wind power deployment is unlikely to affect fisheries, even before the start of construction.

²³ In Appendix B, we also present additional results wherein we aggregate all municipalities at the prefectural level and estimate the weights for coastal fishery production. In addition, we estimate the models at the prefectural level by expanding the sample period from 2004–2018 to 1984–2018 to examine how the length of pre-treatment periods influences our results. For these analyses, we find no statistically significant effects (see Fig. 18).

5.4.2 Leave-One-Out Analysis

The credibility of our results builds on the definition of the donor pools. To analyze the sensitivity of the results to changes in this definition, we eliminate a municipality with a large weight from donor pools and re-estimate the weights. Specifically, we iteratively eliminated one of the five municipalities with the five largest weights. The results are shown in Fig. 8. The solid line measures the difference in fishery production between a treated municipality and its synthetic municipality. The vertical lines with the upper and lower bounds represent 95 % confidence intervals. For ease of visibility, we align the confidence intervals in the same year horizontally with the order of the leftmost results being obtained by eliminating the municipality with the largest weight and the rightmost results with the fifth largest weight.

Figure 8 indicates that our results are robust to changes in the donor pools. The pre-treatment fishery outcome balances reasonably well, and the impacts are statistically indistinguishable from zero in all models. Specifically, the results for Choshi and Kitakyushu are similar to those in Fig. 5a, b of Fig. 5. The results in Goto show that eliminating the municipalities with the fourth- and fifth-largest weights changes the impacts slightly, although the implications do not differ significantly from those of the main analysis.

5.4.3 Excluding Auxiliary Covariates

The analyses thus far include covariates to better approximate pre-treatment trends. One may be concerned that using covariates other than pre-treatment outcomes induces overfitting and poor post-treatment outcome predictions. To address this concern, we exclude covariates from the ASCM estimation. The results are shown in Fig. 9.

Figure 9a–c show that the pre-treatment trends fit quite well in the three municipalities. Notably, the implications that the deployment of offshore wind power exhibits no statistically significant effects do not change in these analyses. Hence, we can conclude that our results are robust to the exclusion of auxiliary covariates. The comparison between Figs. 5 and 9 reveals slight imbalances in the outcome during the pre-treatment period in the models without covariates. Thus, although the exclusion of auxiliary covariates does not affect our results greatly, controlling for them contributes to better approximation of the pre-treatment outcome.

5.5 Effects on Local Per Capita Income

Although this study primarily focuses on negative effects on local fishery production, the SCM approach allows us to measure other economic impacts as well. Offshore wind power deployment is expected to generate positive effects on the local economy mainly through job creation (Varela-Vázquez and del Carmen Sánchez-Carreira 2017; Kahouli and Martin 2018; Kandrot et al. 2020). There will be an increase in job opportunities during the construction and installation phase as well as the operation and maintenance phase. Such an increase will enhance the local demand for non-tradable goods, such as restaurant and real estate, while not necessarily increasing the demand for nationally traded goods (Moretti 2010). Ando (2015) measured the impact of installing nuclear power facilities on

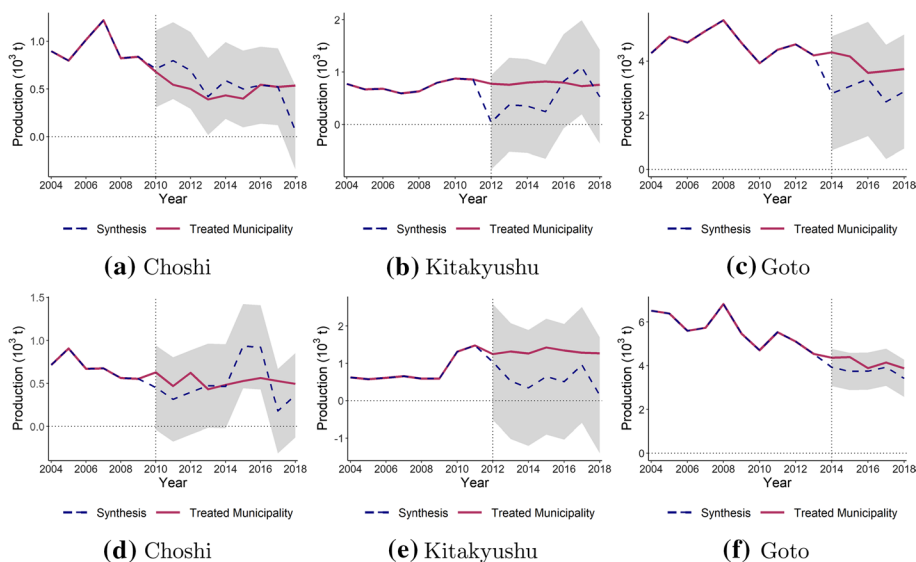


Fig. 6 Spatial spillover for coastal fishery production. Notes: The shaded areas show 95% confidence intervals. **a–c** Show the results of eliminating neighboring municipalities. **d–f** are the results of aggregating neighboring municipalities into a single unit

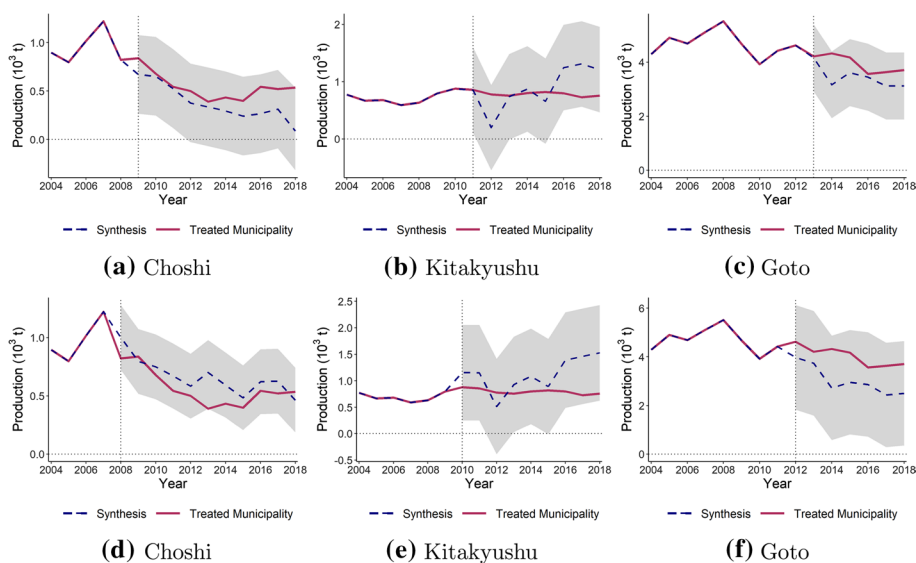


Fig. 7 Backdating analysis for coastal fishery production. Notes: The shaded areas show 95% confidence intervals. **a–c** show the results of the one-year ahead analyses. **d–f** are the results of the two-year ahead analyses

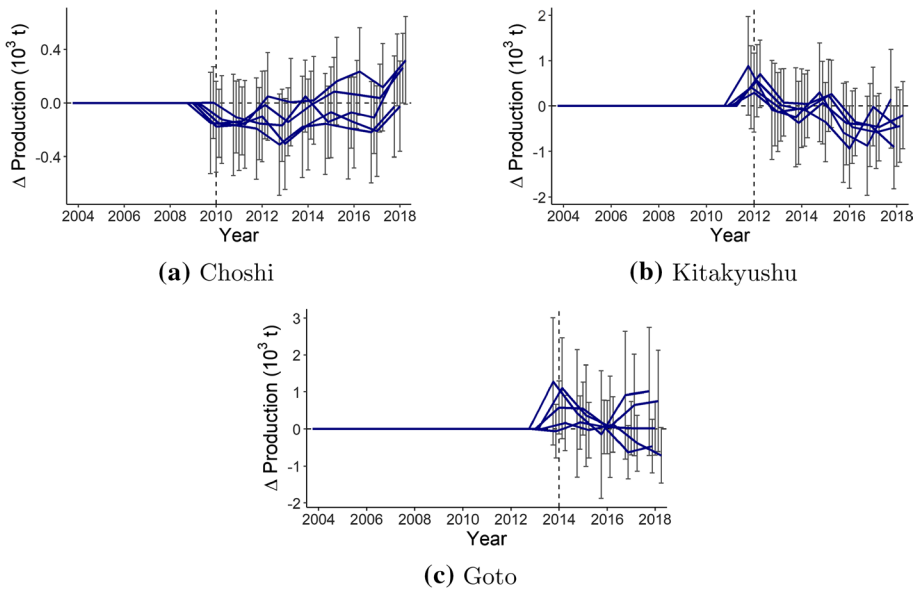


Fig. 8 Leave-one-out analysis for coastal fishery production. Notes: The solid vertical lines with upper and lower bounds show 95% confidence intervals

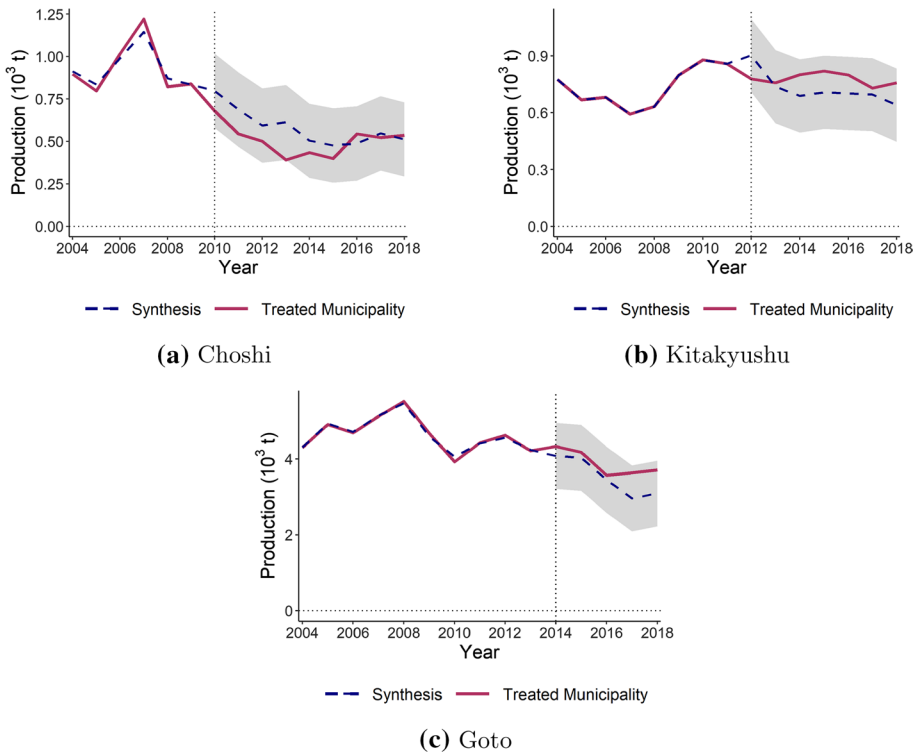


Fig. 9 ASCM results for coastal fishery production without covariates. Notes: The shaded areas show 95% confidence intervals

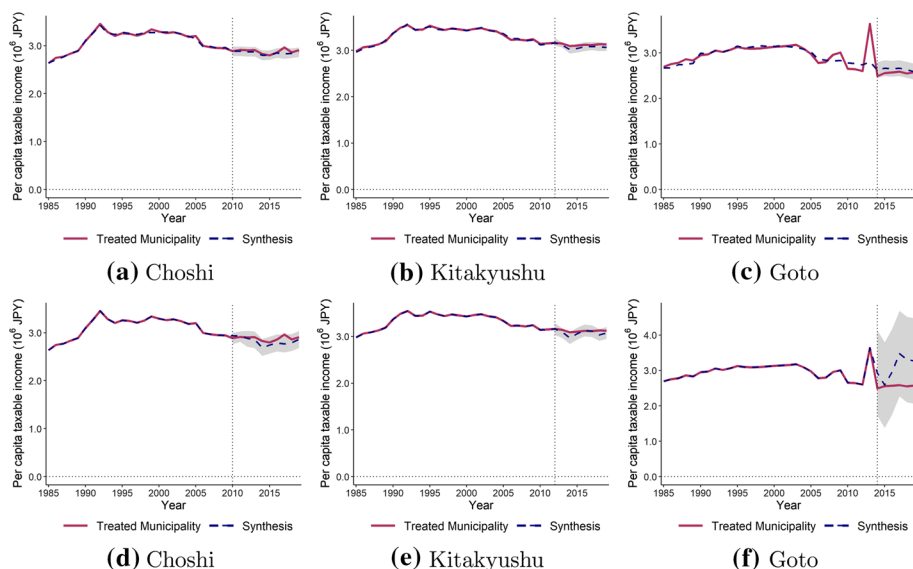


Fig. 10 ASCM estimation results for per capita taxable income. Notes: The shaded areas show 95% confidence intervals. **a–c** show the results without covariates. **d–f** show the results after inclusion of the share of primary sector workers, population, and ratio of elderly people at least 65 years old

local per capita taxable income in Japan and found differential impacts across installed municipalities.

In this subsection, following Ando (2015), we measure the impacts of offshore wind power deployment on local per capita taxable income. We expand the sample range from 1985 to 2019 due to availability of income data. The covariates listed in Table 2 are included, after expanding the sample period, as they would be, if used in Ando (2015). The definition of donor pools is the same as before: the set of coastal municipalities where offshore wind power has not been deployed and whose average per capita income during the pre-treatment period is in the same 10th quantile as that of the treated municipality.²⁴

The results are shown in Fig. 10. Figure 10a–c show the results without the covariates, whereas Fig. 10d–f show the results including them. The pre-treatment trends balance well in all the models, regardless of covariates' inclusion, except for Fig. 10c. Goto experienced a temporary increase in per capita taxable income in 2013, which was due to a large amount of retail stock trade. The jump can be adjusted once we include the covariates (Fig. 10f).

Figure 10b shows that deployment increases per capita taxable income in Kitakyushu two years post-treatment. The increase is 0.071 million JPY (2.13% increase), which is statistically significant at the 5% level, and differences are statistically indistinguishable from zero in other years. Figure 10e shows that once we include covariates, there is no statistically significant difference in any year. The treatment effects in Choshi and Goto are

²⁴ As shown in Fig. 10c, f, Goto experienced a large upward jump in per capita taxable income in a year before the treatment (i.e., 2013), which raises the average during the pre-treatment period. Therefore, we use the values from 1985 to 2012 to calculate the average.

not statistically significant in both models. Thus, we find weak to no evidence of positive economic effects on local per capita taxable income.

6 Discussion

Our results suggest that the three offshore wind power facilities established in Japan are unlikely to disrupt fisheries. Although these farms were installed before the implementation of the PMRE, the results are in line with the requirement of the PMRE that developers of offshore wind power projects should not generate negative impacts on local fisheries. One may think that our results are predictable given the developers' efforts to minimize negative impacts through site selection. However, precise predictions of the ex-post impacts of power facility establishment on fishery production are almost impossible given the lack of a perfect understanding of ecological impacts (Mangi 2013). Thus, our ex-post assessment provides the first and positive evidence regarding fisheries for accelerating the deployment of offshore wind power in Japan. However, the generalization of our results requires caution from at least three perspectives.

First, this study estimates the average treatment effects on the three treated municipalities, unaware of the effects on the other municipalities. If offshore wind power installers selected the three sites anticipating that ecological impacts would be small and fishers would have negligible response to facility installation, the effect of installing offshore wind farms in other sites can potentially be larger. In addition, our results do not imply that facilities at any scale have no impact on fisheries because our estimation builds on three small-scale facilities during the validation phase. It is quite likely, though there is no quantitative causal evidence to the best of our knowledge, the impact on fishery production increases with the scale of offshore wind power facilities (i.e., the number of turbines, their spatial configuration, and power-generating capacity). As discussed in Sect. 2, the METI plans to accelerate the deployment of offshore wind power in Japan in the near future: the GGS plans to attain a capacity of 10 GW by 2030 and at least 30 GW by 2040. The rapid deployment will be realized by installing large-scale wind farms. Further research is required to assess the impacts of large-scale power facilities on fisheries.

Second, the null impacts of offshore wind power deployment might be due to the unit of analysis. Data availability allows us to assess impacts at most the municipal level. If the impacts are much more localized (e.g., at the fishery community level), our estimation results may underestimate them; that is, estimation at a more local level might deliver different results. To conduct such analysis in Japan, one can use the Census of Fisheries, which records detailed fishery information at the fishery community level.²⁵ However, the data are only available every five years. At least two points need to be considered when using the Census of Fisheries. First, the years when data were available might not coincide with the year of construction onset. Second, a longer-period analysis is necessary to balance pre-treatment outcomes well since they can be approximated every five years. Although it is generally difficult to access detailed information on local fisheries over a long period, the investigation of local impacts is an important extension of the current study.

²⁵ See <https://www.e-stat.go.jp/en/statistics/00500210>.

Finally, the lack of investigation of long-term impacts is another limitation of this study. Due to data availability, we measure the impacts after at most eight years since the installation of offshore wind farms (i.e., Choshi). There is no evidence against the possibility that the impacts of offshore wind power facilities on fisheries will grow after a decade of construction. For example, ecosystem changes can take several years to detect (Bailey et al. 2014), affecting fisheries long after installation. Hence, the investigation of longer-term impacts is also an important direction for future work.

Despite these caveats, our results will help offshore wind power installers interested in a moderate-size wind project to better understand the project's possible impacts on the local economy, practically contributing to the ongoing discussion on the better use of coastal regions and marine and maritime areas and coordination between fishery and marine renewable energy facilities (Koundouri 2017; Yates and Bradshaw 2018; Koundouri 2021). Moreover, the approach employed in this study, which applies SCM to observational data, is immediately applicable to the extensions mentioned above. Our approach can provide further policy-relevant insights for policymakers, as the impacts on fisheries can be transformed into monetary values. For example, fish market conditions provide the price of each fish species. By combining this information with the results of the ASCM estimation by species in [Appendix B](#), the monetary impacts on each fish species and the overall impacts on fisheries can be estimated by summing up individual impacts. In addition to the direct monetary impacts on fisheries, our approach allows policymakers to measure indirect monetary impacts on the economy that arise through fish distribution channels. Such information can be utilized for detailed cost-benefit analyses of offshore wind power projects (Snyder and Kaiser 2009).

7 Conclusion

Accelerating the deployment of offshore wind power will be challenging if it disrupts local fisheries. Nonetheless, there is no empirical evidence on whether offshore wind farm installations affect local fisheries. This study fills the gap in the literature by applying SCM, a well-established causal inference technique. With publicly available information on offshore wind farms and panel data on fishery production in Japan, we find no statistically significant effect of offshore wind power deployments on local fisheries. While the findings may sound fascinating for offshore wind power advocates, our results are derived from three small-scale offshore wind farms and may not be generalized to large-scale farms. Moreover, the lack of statistically significant effects does not imply the absence of any impact at all. Therefore, further investigations, especially those focusing on large-scale offshore wind farms, are required, and our SCM approach for impact assessment ensures that it can be applied to studies focusing on large-scale wind farms as well.

Appendix A: Methodological Details

A.1 Objective Function

A.1.1 Synthetic Control Method

Let $\mathbf{X}_0 = (\mathbf{X}_2, \dots, \mathbf{X}_N)$ denote a $T_0 \times N$ matrix of the pre-treatment outcomes of municipalities without offshore wind farms. Similarly, let $\mathbf{Z}_0 = (\mathbf{Z}_2, \dots, \mathbf{Z}_N)$ represent a $K \times N$ matrix of municipality covariates of controlled municipalities. In addition, define $\mathbf{M}_i = (\mathbf{X}'_i, \mathbf{Z}'_i)'$ and $\mathbf{M}_0 = (\mathbf{M}_2, \dots, \mathbf{M}_N)$. Then, the weights of the conventional SCM in Eq. (2), $\mathbf{w}$, can be obtained by solving the following problem:

$$\begin{aligned} & \arg \min_{\mathbf{w}} \|\mathbf{M}_1 - \mathbf{M}_0 \mathbf{w}\|, \\ & \text{s.t. } \forall i \in \{2, \dots, N+1\}, w_i \geq 0 \wedge \sum_{i=2}^{N+1} w_i = 1. \end{aligned} \quad (\text{A.1})$$

The norm in Eq. (A.1) is constructed to account for the importance of each variable as follows:

$$\|\mathbf{M}_1 - \mathbf{M}_0 \mathbf{w}\| = \left(\sum_{k=1}^{T_0+K} v_k \left(M_{k1} - \sum_{i=2}^{N+1} w_i M_{ki} \right)^2 \right)^{1/2},$$

where $\mathbf{v} = (v_1, \dots, v_K)'$ is a vector of weights that reflects the relative importance of each variable. The optimal weight $\mathbf{w}^*$, which is the solution of Eq. (A.1), can be obtained at a fixed value of $\mathbf{v}$ or by updating its value within an algorithm. See Abadie (2021) for a more detailed description of the algorithms.

A.1.2 Augmented Synthetic Control Method

Ben-Michael et al. (2021) propose Ridge ASCM, where weights in Eq. (3) can be obtained as the solution of the following problem:

$$\arg \min_{\eta_0, \boldsymbol{\eta}_x, \boldsymbol{\eta}_z} \frac{1}{2} \sum_{i=2}^{N+1} \left(Y_i - (\eta_0 + \mathbf{X}'_i \boldsymbol{\eta}_x + \mathbf{Z}'_i \boldsymbol{\eta}_z) \right)^2 + \lambda_x \|\boldsymbol{\eta}_x\|_2^2 + \lambda_z \|\boldsymbol{\eta}_z\|_2^2, \quad (\text{A.2})$$

where λ_x and λ_z are hyperparameters that reflect the degree of penalty on extrapolation. When only pre-treatment outcomes are used for approximation, Ben-Michael et al. (2021) show that Eq. (A.2) can be written as

$$\arg \min_{\boldsymbol{\eta} \text{ s.t. } \sum_{i=2}^{N+1} \eta_i = 1} \frac{1}{2\lambda_x} \|\mathbf{X}_1 - \mathbf{X}_0 \boldsymbol{\eta}\|_2^2 + \frac{1}{2} \|\boldsymbol{\eta} - \mathbf{w}\|_2^2, \quad (\text{A.3})$$

and the ASCM estimator of $Y_{it}(0)$ is:

$$\hat{Y}_{1t}(0) = \sum_{i=2}^{N+1} \eta_i Y_{it}.$$

Equations (A.2) and (A.3) allow the weights to take negative values; therefore, extrapolation is allowed.

A.2 Inference

Conventional SCM conducts statistical inference using permutation methods (Abadie et al. 2010). The treatment effects are estimated by iteratively assigning the treatment to units that do not receive it. The placebo effects are then compared to the impact estimated using the unit with the actual treatment. If the unit of interest has a significant impact, it should be larger than the placebo effects.

For statistical inference on the ASCM, Ben-Michael et al. (2021) proposed the conformal inference approach developed by Chernozhukov et al. (2021). The approach first specified a null hypothesis, $\tau_{it} = \tau_0$ (typically $\tau_0 = 0$). Then, the post-treatment outcome was adjusted under the null hypothesis: $\tilde{Y}_{1t} = Y_{1t} - \tau_0$. The ASCM weights $\tilde{\eta}(\tau_0)$ are re-estimated after including $\tilde{Y}_{it}$ in the original data. The p -value is obtained by calculating how the adjusted residual for $t > T_0$, $Y_{1t} - \tau_0 - \sum_{i=2}^{N+1} \tilde{\eta}_i(\tau_0) Y_{it}$, conforms to the residuals for $t \leq T_0$. See Ben-Michael et al. (2021) and Chernozhukov et al. (2021) for more details.

Appendix B: Additional Figures

See Figs. 11, 12, 13, 14, 15, 16, 17 and 18.

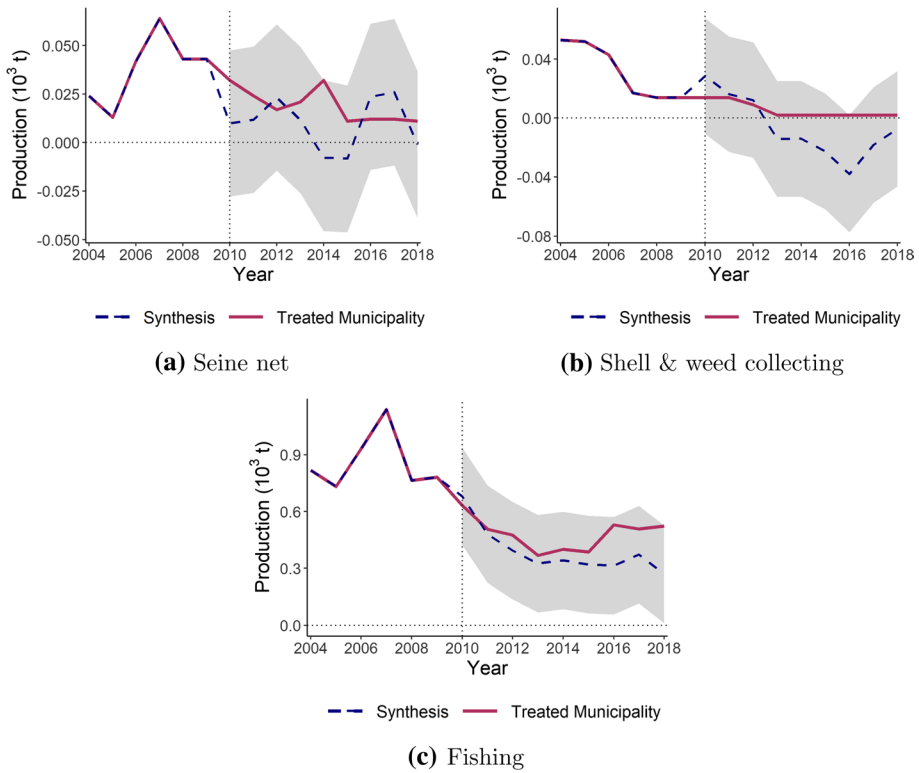


Fig. 11 ASCM estimation results for each fishery type (Choshi). Notes: The shaded areas show 95% confidence intervals

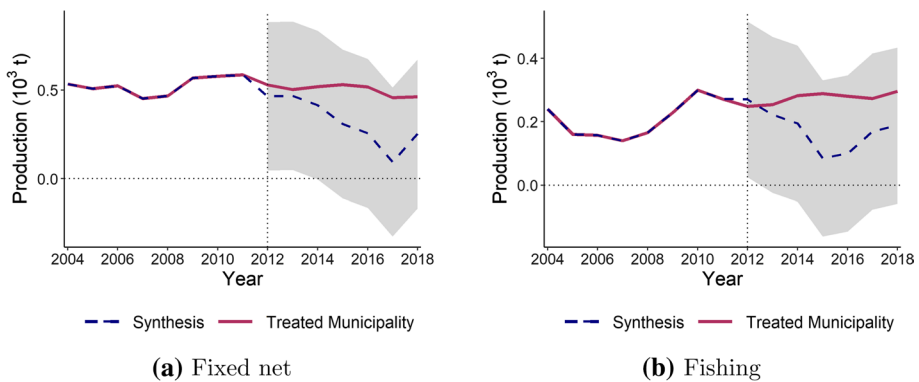


Fig. 12 ASCM estimation results for each fishery type (Kitakyushu). Notes: The shaded areas show 95% confidence intervals

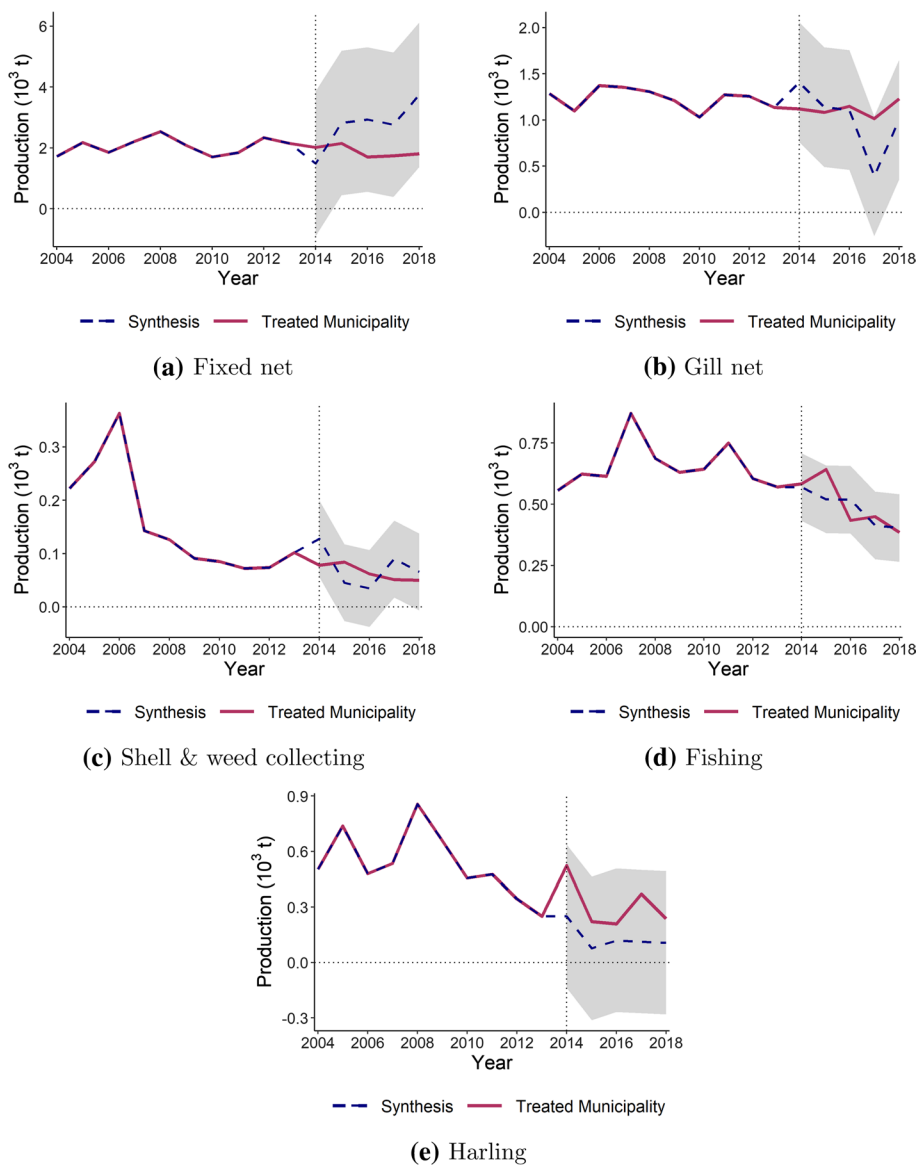


Fig. 13 ASCM estimation results for each fishery type (Goto). Notes: The shaded areas show 95% confidence intervals

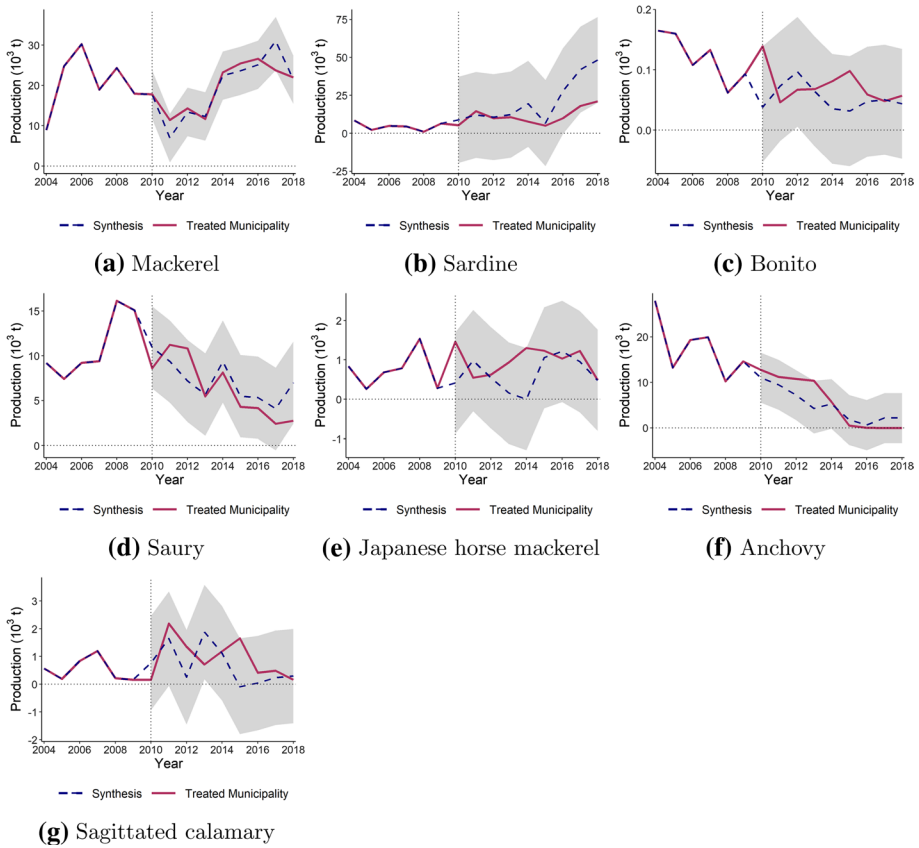


Fig. 14 ASCM estimation results for each fish species (Choshi). Notes: The shaded areas show 95% confidence intervals

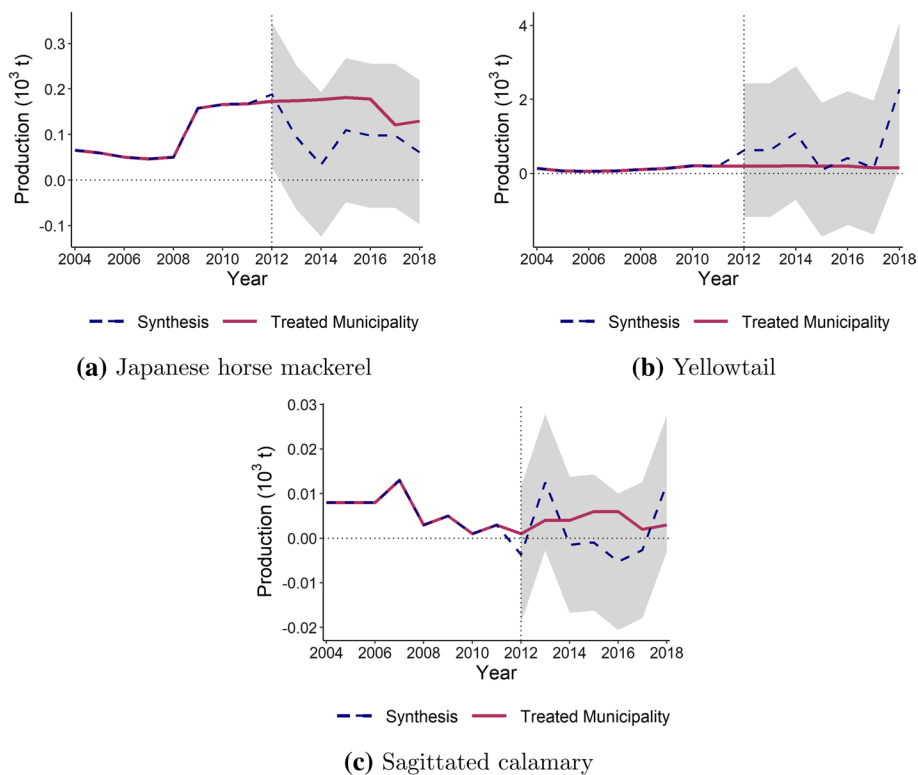


Fig. 15 ASCM estimation results for each fish species (Kitakyushu). Notes: The shaded areas show 95% confidence intervals

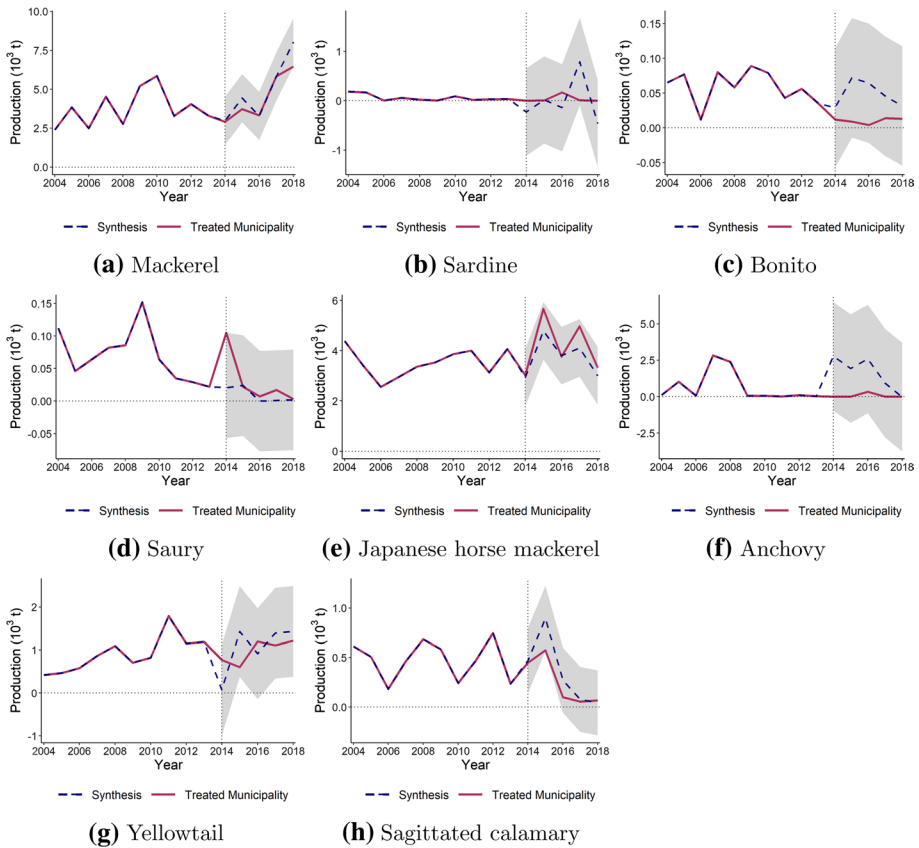


Fig. 16 ASCM estimation results for each fish species (Goto). Notes: The shaded areas show 95% confidence intervals

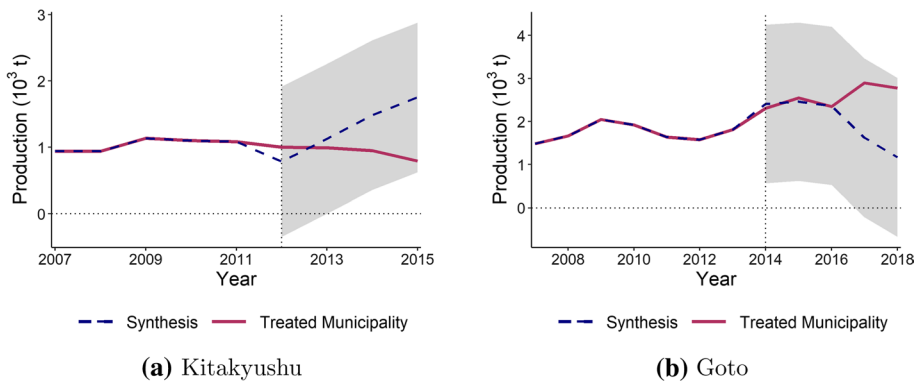


Fig. 17 ASCM estimation results for marine aquacultural production. Notes: The shaded areas show 95% confidence intervals

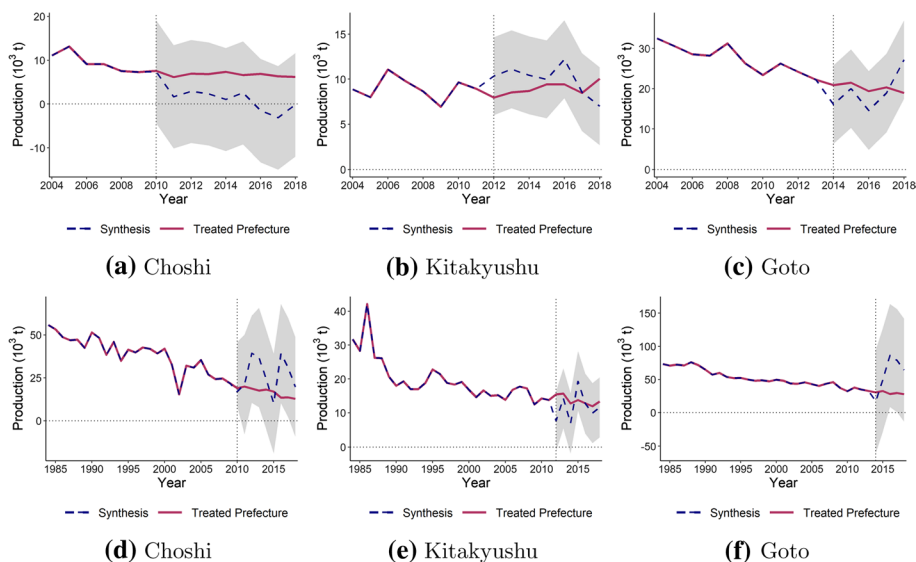


Fig. 18 ASCM estimation results at the prefectural level for coastal fishery production. Notes: The shaded areas show 95% confidence intervals. **a–c** are the results obtained by using the same sample period as in the text. **d–f** represent the results obtained by expanding the sample period from 2004–2018 to 1984–2018. The results in **d–f** were derived using only the shares of fishery types as the covariates

Appendix C: Additional Tables

See Tables 4, 5, 6, 7, 8, 9 and 10.

Table 4 Pooled OLS of total fishery production

	(1)	(2)	(3)	(4)
# fishers	0.010*** (0.002)	0.010*** (0.002)	0.010*** (0.002)	0.010*** (0.002)
# fishery facilities	0.117*** (0.014)	0.117*** (0.014)	0.116*** (0.013)	0.116*** (0.013)
# fishery facility workers	− 0.887*** (0.155)	− 0.889*** (0.155)	− 0.889*** (0.138)	− 0.894*** (0.139)
Production per cooperative ^a	− 8.077 (5.353)	− 8.499 (5.401)	− 1.447 (2.242)	− 2.620 (2.119)
# primary sector workers	− 0.174 (0.212)	− 0.172 (0.216)	− 0.094 (0.216)	− 0.102 (0.217)
Taxable income per capita	1.728 (1.070)	1.344 (1.165)	0.326 (1.342)	− 0.780 (1.825)
Population ^a	1.039 (0.971)	1.040 (0.991)	1.654 (1.072)	1.687 (1.070)
Rate of age ≥ 65	− 10.359* (5.314)	− 12.267* (6.768)	− 8.618 (6.105)	− 12.579 (9.028)
Wind power potential	− 0.688** (0.267)	− 0.684** (0.268)	− 0.584* (0.316)	− 0.578* (0.317)
Mean SST	− 0.942*** (0.176)	− 0.955*** (0.178)	− 1.248** (0.515)	− 1.317** (0.546)
# storm days ^a	− 4.162 (3.164)	− 4.136 (3.172)	3.911 (3.591)	3.827 (3.605)
Share of fixed net	1.588 (2.326)	1.583 (2.325)	3.018 (2.225)	2.976 (2.227)
Share of seine net	− 0.074 (1.961)	− 0.057 (1.959)	0.114 (2.157)	0.070 (2.165)
Share of shell & weed	− 3.428*** (1.230)	− 3.407*** (1.221)	− 1.744 (1.119)	− 1.744 (1.118)
Share of gill net	− 2.015 (2.219)	− 1.994 (2.214)	− 0.975 (2.347)	− 1.027 (2.347)
Share of fishing	0.350 (1.360)	0.359 (1.361)	0.283 (1.130)	0.273 (1.132)
Share of harling	− 2.359 (1.960)	− 2.326 (1.957)	− 2.965* (1.752)	− 2.927* (1.753)
Adjusted R^2	0.611	0.611	0.647	0.648
Year fixed effects		✓		✓
Prefectural fixed effects			✓	✓

This table shows the results of regressing total fishery production on municipal and prefectural characteristics with and without year and prefectural fixed effects. Standard errors in parentheses are clustered at the municipality level. The number of observations was 8235, with 549 municipalities. ^acoefficients multiplied by 1000 for rescaling. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 5 Pooled OLS of coastal fishery production

	(1)	(2)	(3)	(4)
# fishers	0.003** (0.001)	0.003** (0.001)	0.003*** (0.001)	0.003*** (0.001)
# fishery facilities	− 0.007*** (0.003)	− 0.007*** (0.003)	− 0.006** (0.003)	− 0.006** (0.002)
# fishery facility workers	0.108 (0.077)	0.108 (0.077)	0.095 (0.081)	0.094 (0.080)
Production per cooperative	− 1.616 (1.298)	− 1.686 (1.317)	− 0.456 (0.508)	− 0.816 (0.519)
# primary sector workers	− 0.140 (0.088)	− 0.139 (0.088)	− 0.176** (0.085)	− 0.176** (0.085)
Taxable income per capita	0.379 (0.250)	0.359 (0.253)	− 0.231 (0.371)	− 0.462 (0.449)
Population	− 0.323 (0.290)	− 0.328 (0.294)	0.143 (0.282)	0.137 (0.282)
Rate of age ≥ 65	− 3.288* (1.827)	− 3.433* (2.050)	− 4.191** (1.778)	− 5.159** (2.279)
Wind power potential	0.014 (0.109)	0.014 (0.110)	0.098 (0.113)	0.102 (0.114)
Mean SST	− 0.189*** (0.046)	− 0.191*** (0.047)	− 0.211* (0.114)	− 0.227* (0.123)
# storm days	− 0.402 (0.732)	− 0.387 (0.734)	0.467 (0.828)	0.441 (0.829)
Share of fixed net	2.517*** (0.504)	2.514*** (0.504)	2.855*** (0.521)	2.845*** (0.521)
Share of seine net	1.867*** (0.553)	1.869*** (0.554)	2.042*** (0.453)	2.030*** (0.455)
Share of shell & weed	0.192 (0.219)	0.192 (0.220)	0.424* (0.253)	0.425* (0.255)
Share of gill net	0.447 (0.413)	0.448 (0.413)	0.429 (0.522)	0.418 (0.522)
Share of fishing	0.294 (0.179)	0.298* (0.180)	0.167 (0.200)	0.167 (0.202)
Share of harling	0.958***	0.967***	0.541*	0.558*
Adjusted R^2	0.360	0.360	0.434	0.434
Year fixed effects		✓		✓
Prefectural fixed effects			✓	✓

This table shows the results of regressing coastal fishery production on municipal and prefectural characteristics with and without year and prefectural fixed effects. Standard errors in parentheses are clustered at the municipality level. The number of observations was 8235, with 549 municipalities. ^acoefficients multiplied by 1000 for rescaling. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 6 Difference-in-difference estimation of total fishery production

	Choshi		Kitakyushu		Goto	
	(1)	(2)	(3)	(4)	(5)	(6)
τ	- 1.265 (2.042)	1.341 (3.250)	0.288*** (0.105)	0.101 (0.177)	0.363 (0.434)	1.082 (1.322)
Covariates		✓		✓		✓
Observations	735	735	780	780	765	765
Municipalities	49	49	52	52	51	51
Adjusted R^2	0.180	0.257	0.329	0.340	0.172	0.221

This table presents the results of the DID estimation by municipalities with and without covariates. The outcome is the total fishery production. Municipalities neighboring treated municipalities are excluded. The standard errors in parentheses are clustered at the municipality level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 7 Difference-in-difference estimation of coastal fishery production

	Choshi		Kitakyushu		Goto	
	(1)	(2)	(3)	(4)	(5)	(6)
τ	- 0.318*** (0.031)	- 0.355*** (0.064)	0.151*** (0.028)	0.279* (0.159)	1.136** (0.523)	1.575** (0.692)
Covariates		✓		✓		✓
Observations	735	735	750	750	765	765
Municipalities	49	49	50	50	51	51
Adjusted R^2	0.089	0.190	0.082	0.105	0.149	0.220

This table presents the results of the DID estimation by municipalities with and without covariates. The outcome is coastal fishery production. Municipalities neighboring treated municipalities are excluded. The standard errors in parentheses are clustered at the municipality level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 8 Productions of major fish species by fishery types

	Coastal	Non-coastal	Coastal share (%)
Mackerel	47.02	494.96	8.68
Sardine	130.40	391.98	24.96
Scallop	0.01	304.76	0.00
Bonito	0.62	247.09	0.25
Pollack	5.13	122.37	4.02
Saury	0.33	128.60	0.26
Japanese horse mackerel	19.74	98.02	16.76
Anchovy	53.99	57.23	48.55
Yellowtail	51.71	48.25	51.73
Salmon	81.53	2.43	97.11
Pacific cod	3.48	47.18	6.87
Sagittated calamary	5.89	41.82	12.34

This table presents the total production of each major fish species in Japan categorized by two fishery types: coastal and non-coastal fisheries. The third column represents the ratio of coastal fishery production to total production

Table 9 Aquacultural production

	Choshi		Kitakyushu		Goto		Others	
	Mean	S.D.	Mean	S.D.	Mean	S.D.	Mean	S.D.
Production (10 ³ t)	0	0	0.99	0.10	2.09	0.48	1.81	5.45
Observation	0		9		12		5858	

This table presents summary statistics of aquaculture production by municipalities with and without off-shore wind farms. Data from Kitakyushu from 2016 to 2018 were unavailable due to data confidentiality. “Others” exclude observations whose production was not available due to data confidentiality

Table 10 Comparison of covariates over treatment and control groups

	Choshi		Kitakyushu		Goto		Control
	Treated	Synthesis	Treated	Synthesis	Treated	Synthesis	
<i>Production by fishery types</i>							
Fixed net	0.00	0.83	×	×	2.01	1.62	0.67
Seine net	0.03	0.01	×	×	×	×	0.34
Gill net	×	×	×	×	1.19	0.82	0.29
Shell & weed	0.02	0.03	0.23	0.20	0.12	−1.35	0.24
Fishing	0.63	0.48	0.52	0.39	0.60	0.65	0.06
Harling	×	×	0.24	0.08	0.46	0.76	0.03
<i>Covariates</i>							
Production per cooperative	61.15	82.69	7.33	18.25	25.06	23.38	40.75
Taxable income per capita	1.18	1.19	1.23	1.3	0.87	0.88	1.06
Mean SST	19.57	19.45	19.29	19.25	21.38	21.39	18.27
# fishers (10 ³)	0.48	0.49	0.69	0.58	1.35	1.33	0.31
Rate of age >= 65	0.29	0.29	0.25	0.25	0.33	0.34	0.29

This table summarizes each treated municipality’s fishery productions and covariates, its corresponding synthesis, and all controlled municipalities averaged over the sample periods. # fishers and Rate of age ≥ 65 are observable every five years, whereas the other variables are observed every year (i.e., 2004–2018). “×” represents missing values due to data confidentiality

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